UF Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

Lecture 1B: Applications of Human Factors and Classification of Ergonomic Problems

EIN 3241: Human Factors and Ergonomics
Spring 2026

January 14, 2026

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Admin Issues

- Drop/Add: January 12 – 16
- Final Exam Rescheduled: Tuesday 04/28/2026 @ 8:00 PM - 10:00 PM
- DRC Quizzes

What is "Human Factors and Ergonomics"?

From the Human Factors and Ergonomics Society (HFES):

- Ergonomics (or human factors) is the scientific discipline concerned with the understanding of **interactions** among **humans** and other **elements of a system**, and the profession that applies theory, principles, data and methods to design in order **to optimize human well-being and overall system performance**.
- Ergonomists contribute to the design and evaluation of tasks, jobs, products, environments and systems to make them compatible with the needs, abilities and limitations of people.

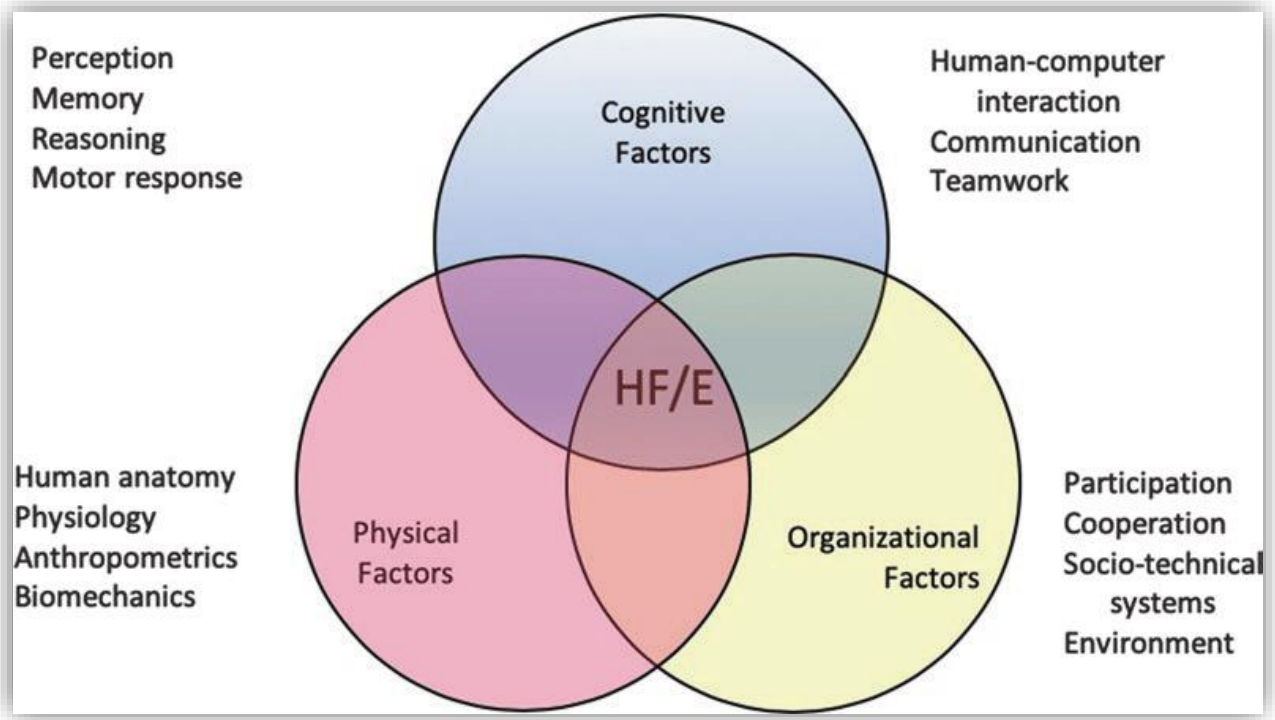
HFES
Systems That Work for Humans

https://www.hfes.org/about-hfes/what-is-human-factorsergonomics

Human Factors VS Ergonomics

- Terms are synonymous.
  - “Human factors” typically used in U.S.
  - “Ergonomics” typically used in elsewhere.

- Some have attempted to differentiate terminology:
  - Human factors – Focused on human cognition, decision making, and mental demands.
  - Ergonomics – Focused on physical demands and physical workload responses.

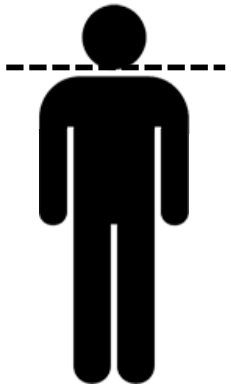


Why HF/E is unlike other disciplines?

- The objects of analysis (humans) exhibit immense variability
- Design decisions are often based on scientific principles as opposed to scientific laws
- HF is an inter-disciplinary field

“From the Neck Up” or “From the Neck Down”

Brain or Central
Nervous System
or Mind

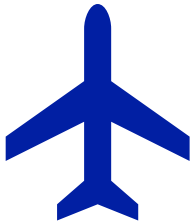


Musculoskeletal,
Cardiovasular,
Respiratory, and
Nervous systems

Why Should We Care… AGAIN?!

- Human behavior is major contributor to failures in safety critical systems

- 44,000 to 98,000 deaths annually due to preventable medical errors
- At least 1.5M injuries a year

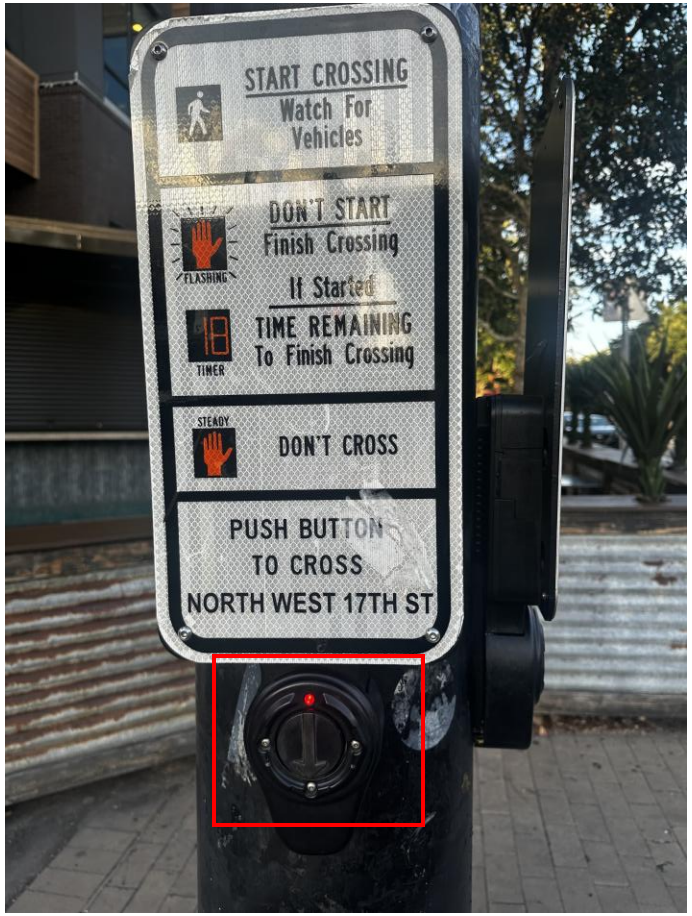


From 2012–2021, pilot error contributed to 56% of fatal accidents

95% of all roadway crashes

Isn't this just intuitive, why study it?

- We designed it for the average…
- People are too variable…
- "Can't people read the instructions?"



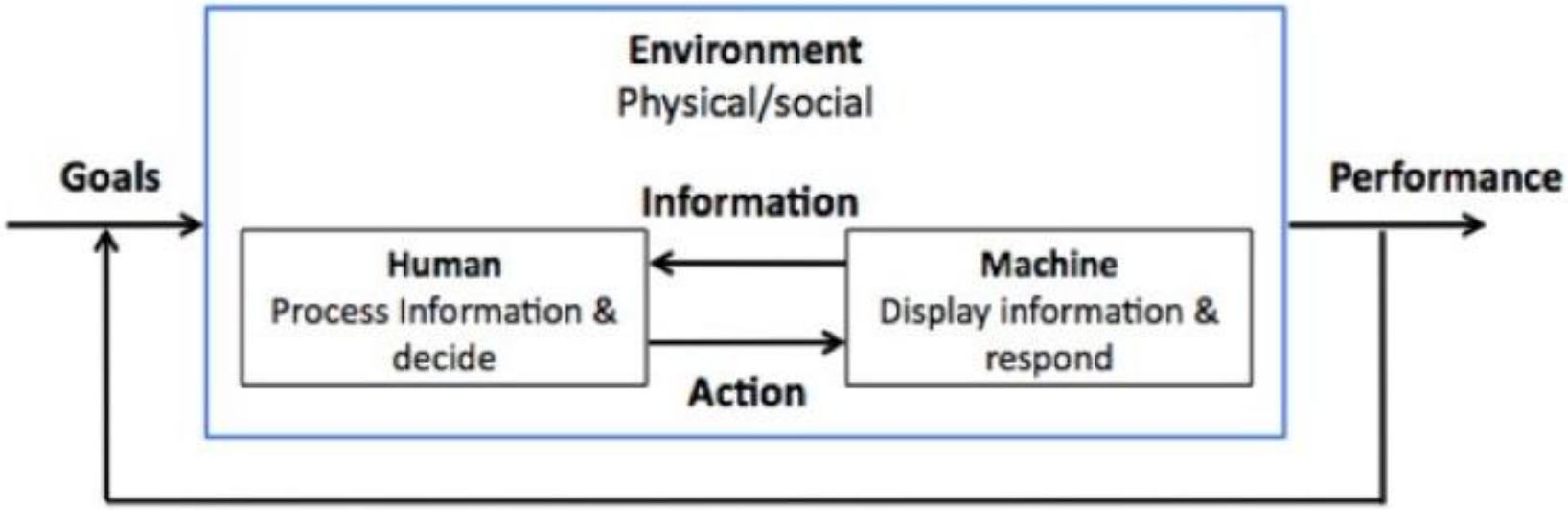
On Campus….

Motivations for studying HF/E

- Human error ≠ cause of system failure
- Complex system designs may not be compatible with humans (Wickens, 1993)
- Example of a complex system: Boeing 737 Max 8 (2018-2019)
  - Changes in aircraft design and underlying systems led to two major aviation accidents: Ethiopian Airlines Flight 302 (157 fatalities), Lion Air Flight 610 (189 fatalities)
  - Flight dynamics changed, Automation (MCAS) used to compensate
  - Training issues
  - Data overload

- Need for safety in complex sociotechnical systems
- Usability in simple technologies

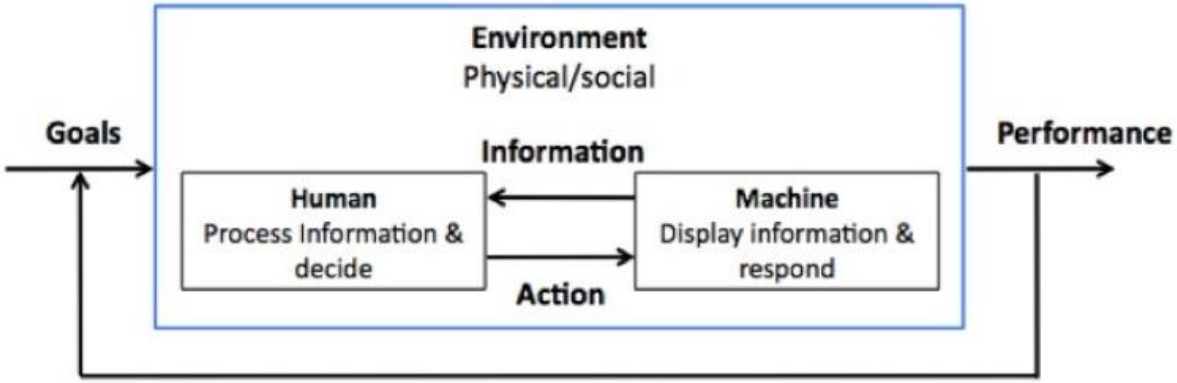
Basic Ergonomic model



- The Ergonomic model is also known as the human-in-the-loop model.
- The human-machine system includes one or more humans and machines (or tools) in a given environment performing one or more tasks to achieve one or more goals.
- The human-computer interaction includes interfaces between human and computers.

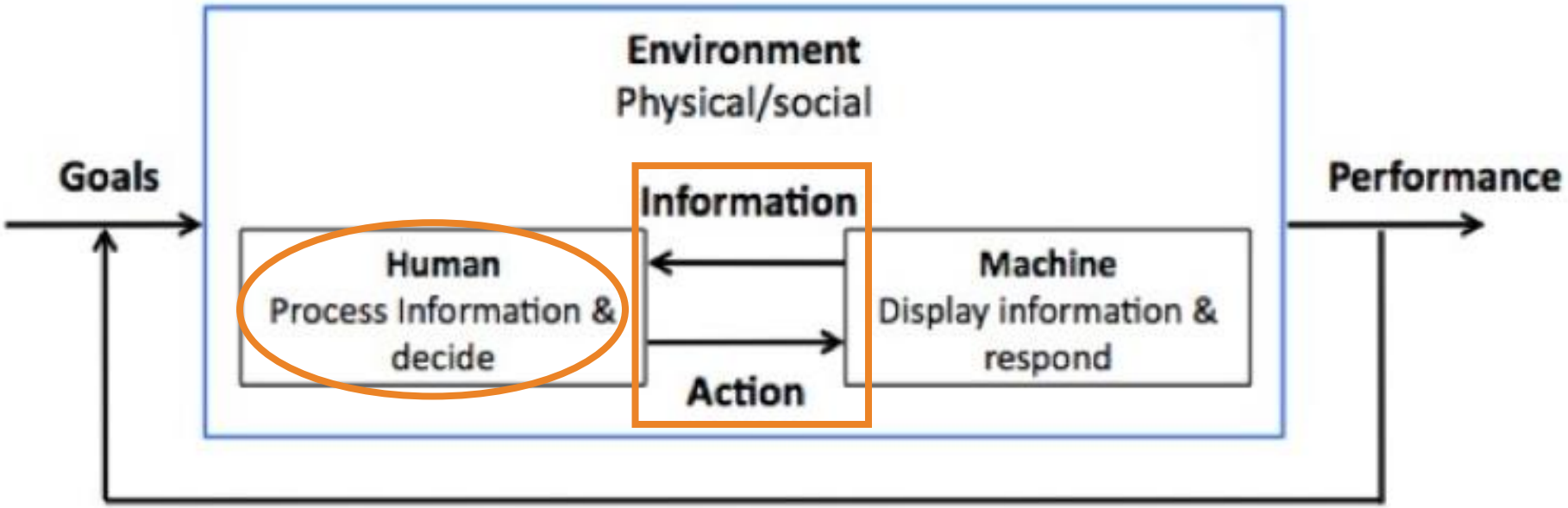
Goals & Standards

- Goals should describe:
  - The reason(s) that the system exist
  - Why the task is performed -- not how.
  - The expected work/task performance
    - There may be multiple goals
    - There may be short-term and long-term goals
    - Workers, employers, and consumers all may have different goals
    - Goals may be explicitly stated or implied



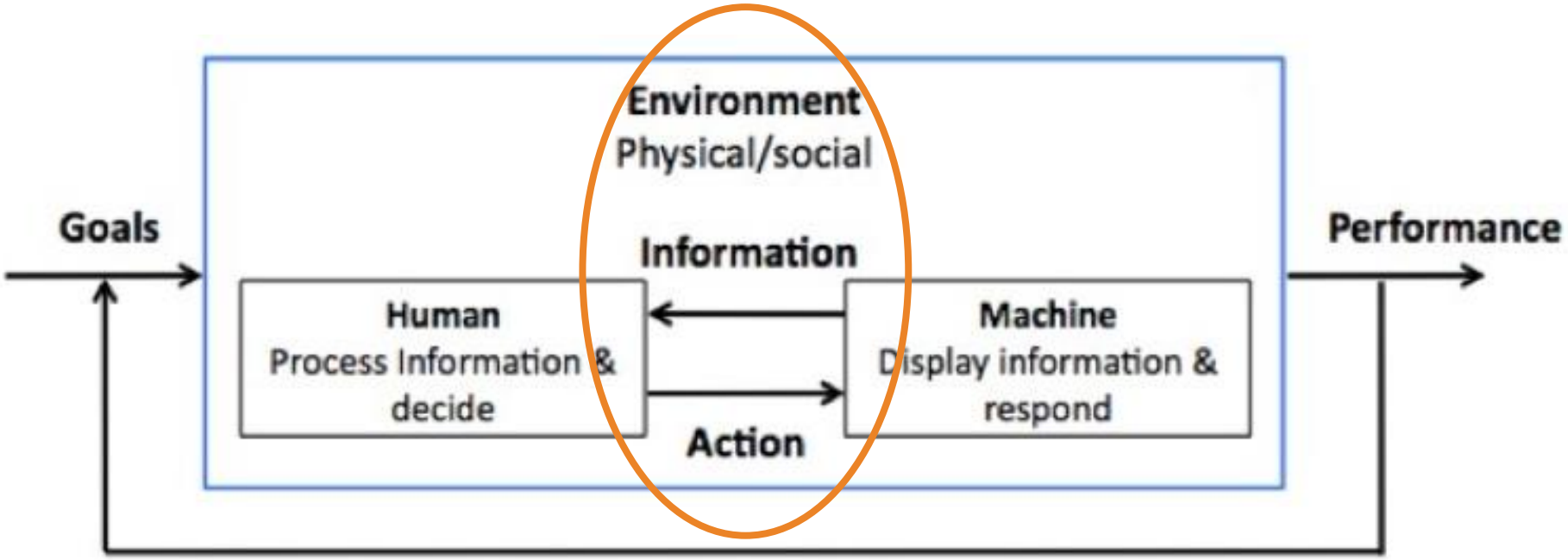
- Work standards describe:
  - Quantity, quality, and time for achieving the goals
  - Standard is set by an employer, equipment manufacturer, and worker/user
    - Methods, equipment/tools, workspace, environment, Materials
    - Worker/user qualifications, training, special needs

Basic Ergonomic model



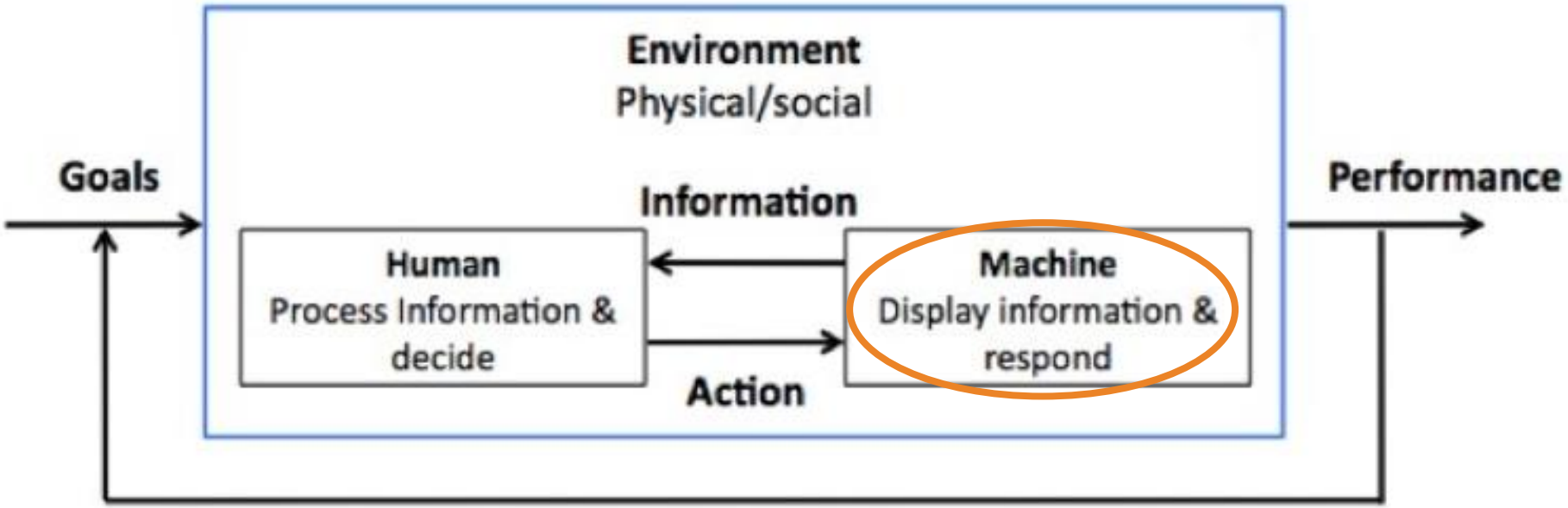
- The **human** part of the system gathers and processes information and acts to maintain control of the system and achieve the desired system goals.
- The gathering, processing of information and resulting actions can be characterized as serial and parallel tasks.

Basic Ergonomic model



- Physical environmental factors can interfere with gathering information, control actions and physical capacities.
- Psycho/social factors affect processing information and decision making

Basic Ergonomic model

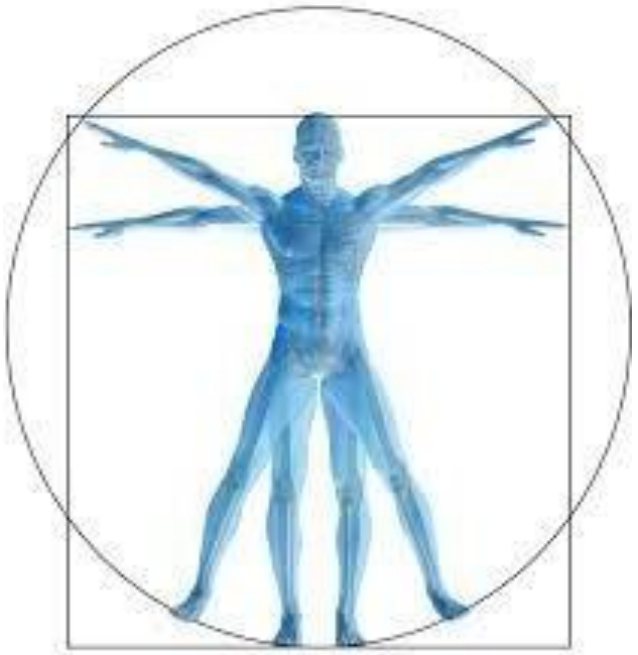


"**Machine**" broadly refers to objects that human interact with to achieve a desired goal

Applications of Human Factors and Ergonomics

HF/E continues to be successfully applied in various areas:

- Aerospace
- Aging
- Healthcare
- Usability
- Product design
- Surface transportation
- Office design
- etc…



In-Class Activity!

- In small groups, choose a couple domains and come up with real-world examples of how HF/E principles might be applied in such areas.
- Think about:
  - A product, interface, or system
  - A safety or efficiency challenge
  - How HF/E could help solve it
- You'll have 5 minutes. Then we'll share examples with the class!
- HF/E continues to be successfully applied in various areas: Aerospace, Aging, Healthcare, Usability, Product design, Surface transportation, Office design…
- You are welcome to also think about other unmentioned application domains!

Applications of Human Factors: Aerospace

- Cockpit design
- Workload Management
- Training Design
- Space Missions

Applications of Human Factors: Aging

- E.g., designing phones for different age groups

Applications of Human Factors: Healthcare

- Medical Device Design
- Medication Safety
- Operating Room Ergonomics
- Electronic Health Records (EHRs; MyChart)
- Emergency Department Flow
- Patient Handling and Lifting
- ....

Applications of Human Factors: Usability

- Website Design
- App Design
- Phone Interface
- Voice Assistants
- ....

Applications of Human Factors: Product Design

Example: Conventional vs. Ergonomic Valve Wrench (Al-Qaisi & Aghazadeh, 2018)

- Conventional:
  - Straight handle with uniform grip
  - Often requires awkward wrist angles and high force

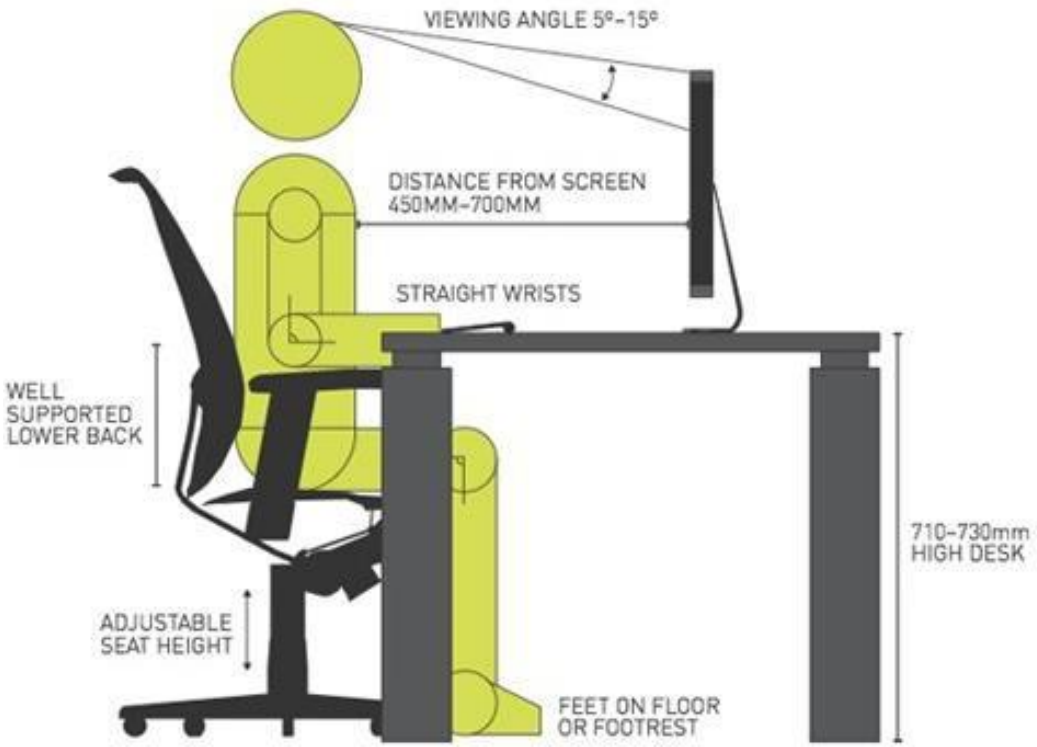
- Re-Designed:
  - Adjustable between 0 and 180 degrees
  - Reduces muscle strain and risk of injury

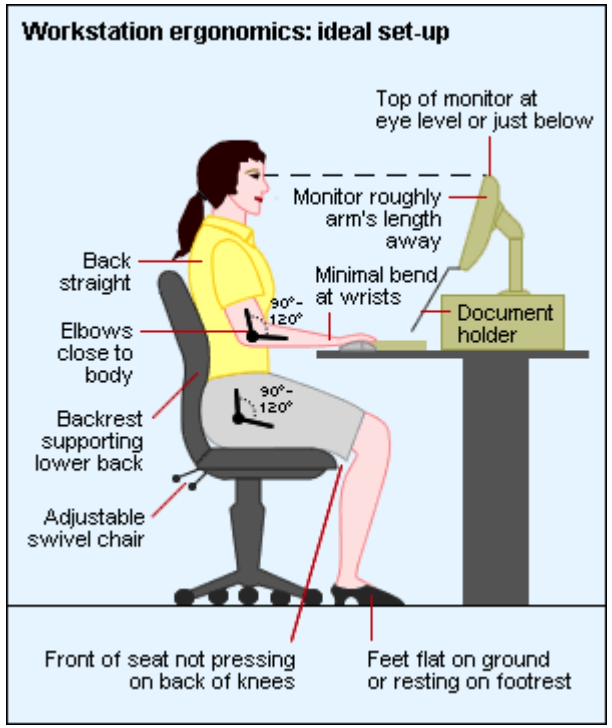
Applications of Human Factors: Surface Transportation

- Dashboard…
- Traffic Signal Timing
- Road Signs
- Highway Signage Design
- …

Applications of Human Factors: Office Design

- Ergonomic Workstations
- Lighting
- Layout
- …





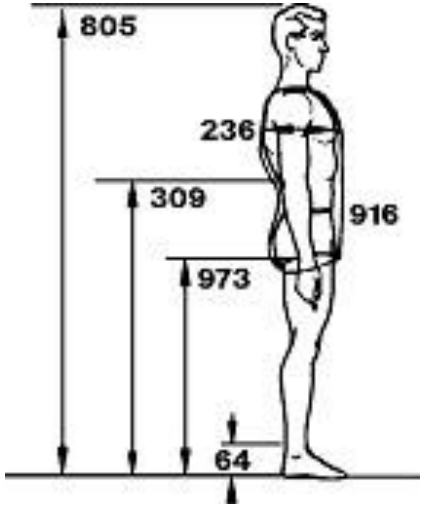
Classification of Ergonomic Problems

- Ergonomic problems can be classified into the following broad (and sometimes overlapping) categories:

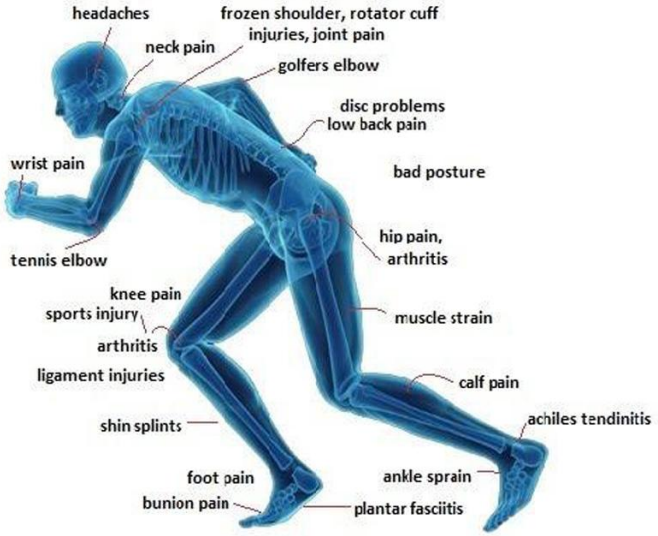
1. Anthropometric
2. Musculoskeletal
3. Cardiovascular
4. Cognitive
5. Psychomotor
6. Organizational

Classification of Ergonomic Problems

1. Anthropometric
   - Study of human body dimensions and body proportions. Joints' range of motion and strength norms are also considered in anthropometrics
   - E.g., reach distances, sitting eye height, also includes weight and volume
   - Application can reduce the dimensional conflict between functional space geometry and the human body



2. Musculoskeletal
   - Tasks that strain the muscular and skeletal systems are in this category
   - E.g., overexertion injuries, posture analysis, repetitive tasks



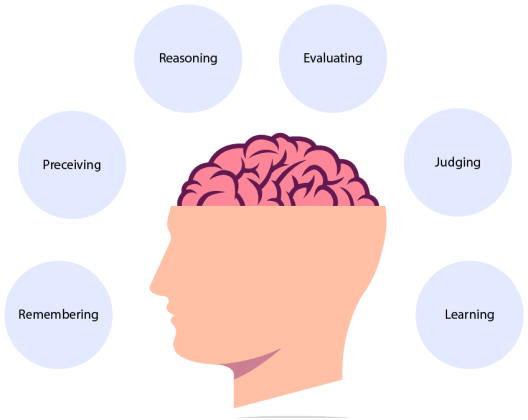
Classification of Ergonomic Problems

3. Cardiovascular

- Tasks that stress the heart and circulatory system
- E.g., heart rate, blood pressure, oxygen consumption, energy expenditure
- When a demand is placed on the cardiovascular system, the heart pumps more blood to the muscles to meet the elevated oxygen requirements.
- If the activity is excessive, it can lead to fatigue, exhaustion, dehydration, and produce both short- and long-term negative effects

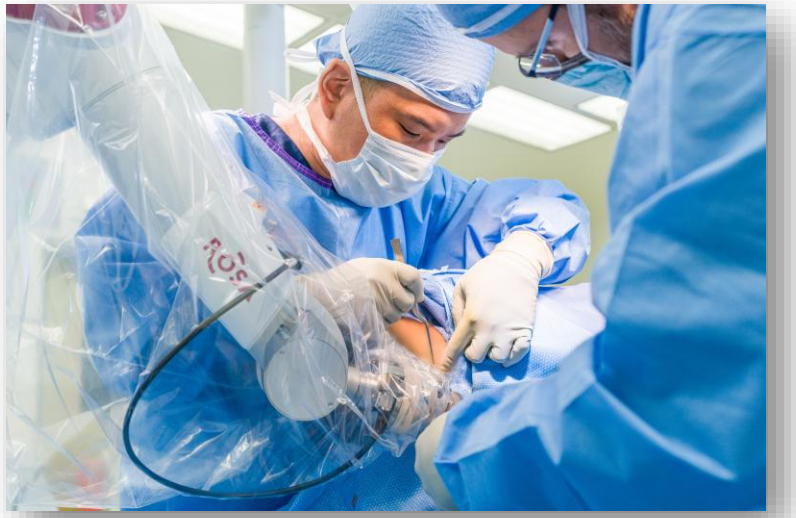
4. Cognitive

- Related to either information-processing overload or underload.
- These functions should be sufficiently utilized for the maintenance of an optimum state of awareness
- Tasks should be designed to be compatible with cognitive capabilities and to complement human functions with machine functions for enhanced performance

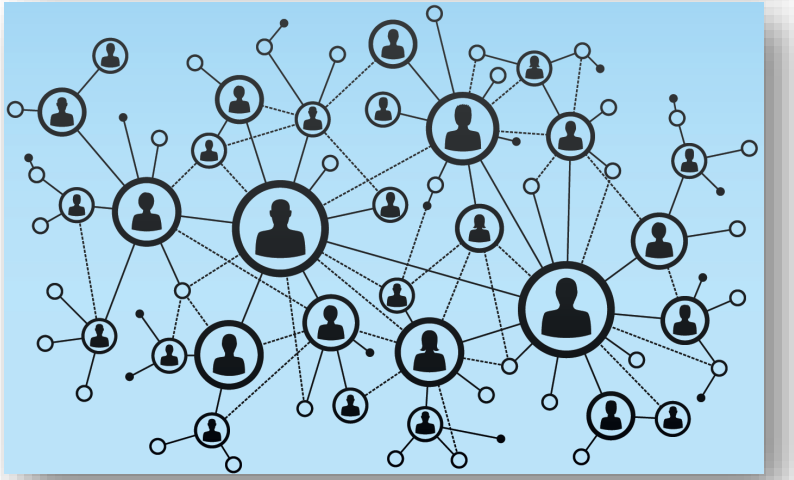


Classification of Ergonomic Problems

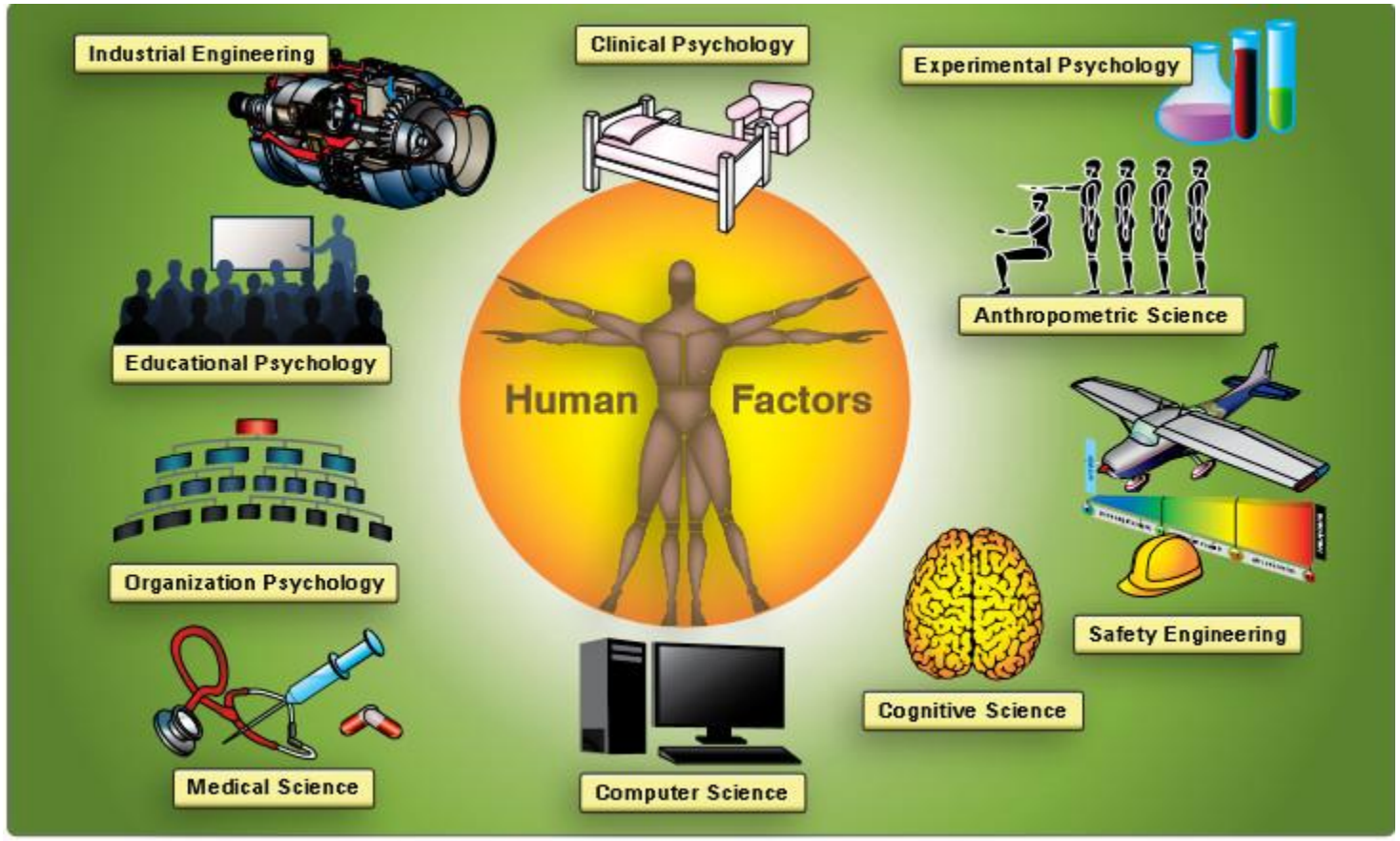
5. Psychomotor
   - Problems that strain the systems that respond to visual or auditory stimuli with movement or muscular activity.
   - E.g., paced manipulative work with significant visual demands, surgical procedure, sewing task



6. Organizational
   - Problems dealing with the organizational structures, policies, and processes in the work environment.
   - E.g., design of shift hours, communication, job satisfaction, motivation, supervision, teamwork, ethics, and culture



Human Factors & Ergonomics is an Interdisciplinary Field



What Did We Learn Today?

- Defined HFE AGAIN!
- Basic Ergonomics Model
- Human Factors Applications
- Classification of Ergonomic Problems

Aerospace
Aging
Healthcare
Usability
Product Design
Transportation
Office Design

And More…

Anthropometric
Musculoskeletal
Cardiovascular
Cognitive
Psychomotor
Organizational

Next Time…

- History of the field!
- Organizations/Societies
- Legislative and Regulatory Issues Impacting Ergonomics
  - OSHA
  - NIOSH
  - DoD
  - ISO
  - …
- Justifying an Ergonomic Program!!!!

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LEADING THE CHARGE, CHARGING AHEAD